

Bitácora — Compositional Depth Research Line

Live experimental log, breakthrough attempt

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TL;DR del día

Breakthrough candidato confirmado: TreeMoTE (recursive shared-cell architecture) achieves **100% accuracy at test depths up to $d=8$ OOD** on balanced RPN mod-7, where Transformer/MoTE/StackMoTE plateau at 17-25% under identical conditions.

Implicación: la ley empírica α (decay exponencial con depth) no es un límite fundamental — mide el **costo de dispatch discovery**. Arch que no tiene que aprender dispatch (tree-recursive) $\rightarrow \alpha \rightarrow 0$.

Estado: todos los stress tests críticos completados. Multi-seed 3/3 confirmed. Depth extrapolation to $d = 12$ works (94–100%). Coverage hypothesis refined via $p = 31$ test.

1 Cronología condensada

2026-04-23 (día 1)

- **Planteo inicial:** MoBE-BCN perdió vs Transformer matched-params en math LM (4.5% PPL gap sobre 11 evals). Arch claim muerto en math LM.
- **Pivot:** Investigar compositional depth extrapolation en MoTE/StackMoTE.
- **E1 (dispatch probe):** StackMoTE controller aprende `digit→push` (91.5%) / `op→pop_apply` (64%) sin supervisión. H1 (discovery) muerta.
- **E2 (balanced RPN single-depth):** Todas las archs $\approx 70\%$ training, chance OOD. Single-depth oculta diferencias.
- **E3 (hard stack STE):** +2-4pp modesto. H2 parcial.
- **A (residual ablation):** Sin residual +20pp training fit, 0pp OOD. H4 parcial.
- **B (operator isolated):** `Op_bil` grokea mod-7 en 500 steps aislado. Primitivos son expresivos. H3 muerta.
- **C1 (curriculum {2,3}):** StackMoTE $d4=25\%$, Trans $d4=16\%$ — primer OOD real.
- **C2 (curriculum {2,3,4}):** Patrón estable.
- **Multi-seed C3:** Stack $d5=22\%\pm0.4\%$, Trans $=15\%\pm0.9\%$. Robusto.

2026-04-24 (día 2) — mañana

- **Scale sweep (6 configs, 31× params):** OOD d5 = 22%±0.8% invariante. Training fit escala normal.
- **Research agent audit:** descubre que baseline era mal calibrado — balanced RPN mod-7 tiene label=0 al 20% por × colapsando. Chance real 14.3%, majority-class 20%.
- **Label-balanced multi-seed:** Bajo null model correcto, Stack +6.8pp, Trans +3.2pp. Gap ~ 2× (no 3-5×). Finding sobrevive reducido.
- **Extended 50K retraction:** d128_L2 outlier no se reproduce con extended training. Steady state d5=23%. Single-seed noise.

2026-04-24 — tarde: descubrimiento de la ley α

- **Data mining 42 runs:** descubro ley empírica

$$\text{acc}(d) = \text{floor} + (\text{acc}(d_{\min}) - \text{floor}) \cdot e^{-\alpha(d-d_{\min})}$$

- **Literature search agent:** forma exponencial es folklore (LeCun 2023, “Beyond Exponential Decay” 2505.24187). Nuestra contribución: validación empírica + α como arquitectural fingerprint + leave-one-out validation.
- **Multi-seed α :** Stack $\alpha=0.42\pm0.09$, MoTE= 0.76 ± 0.05 , Trans= 0.78 ± 0.18 . Ordering robusto.
- **2-param model (floor + α):** R^2 0.795→0.826. Stack floor $\approx 2\times$ Trans floor.
- **Alpha law prediction test:** fit $d=\{2,3,4,5\}$, predict $d=6,7,8 \rightarrow$ MAE 1.7-3.2pp. La ley extrapola fuera del rango fiteado.
- **Param harvest + entropy analysis:** Dispatch entropy correlates con α across 3 curricula. α depende de H .
- **Paper v4 compilado** — paper/compositional_depth_law.pdf.

2026-04-24 — late tarde: BREAKTHROUGH

- **TreeMoTE implementation:** recursive shared-cell architecture. No dispatch controller — estructura del árbol explícita in the computational graph.
- **First TreeMoTE run:** $d=2..7 = 100\%$, $d=8 = 99.4\%$.
- **Hypothesis confirmed:** α (decay rate) = dispatch tax. Arch sin dispatch $\rightarrow \alpha = 0$.
- **TreeMoTE vs Stack vs Transformer head-to-head:**

arch	d=2 (train)	d=5 (OOD)	d=8 (OOD)
TreeMoTE	1.00	1.00	0.996
Stack_noresid	0.47	0.24	0.20
Transformer	0.47	0.23	0.17

Gap +80pp a $d=8$.

2 Stress tests (2026-04-24 PM)

Test 1: Multi-seed (3 seeds)

seed	d2	d3	d4	d5	d6	d7	d8
42	1.00	1.00	1.00	1.00	1.00	1.00	0.994
123	1.00	1.00	1.00	1.00	1.00	1.00	0.998
7	1.00	1.00	1.00	1.00	0.998	0.996	0.984

3/3 CONFIRMED. TreeMoTE es robust across seeds.

Test 2: Depth d=12 extrapolation

Train {2,3,4,5}, test d hasta 12. 7 depths beyond training:

	d6	d7	d8	d9	d10	d11	d12
TreeMoTE	0.998	1.000	0.996	0.986	0.980	0.944	0.960

Perfecto hasta d=12. Decay efectivo $\alpha \approx 0.01-0.02$ (vs Transformer $\alpha \approx 0.78$).

Test 3: Low-data (n=100 per depth)

Memorizó training (CE=0) pero no generalizó (accuracy $\sim 17\%$ all depths). Sin data suficiente (3% coverage), cell no aprende operator.

Cell table audit (MECHANISTIC PROOF)

Extract trained cell, apply to all (a, b, op) triples at $p = 7$. Result: **147/147 = 100%**. Cell aprendió la tabla mod-7 exacta. Depth extrapolation es consecuencia: cell exacta $\times$ recursión estructural = 100%.

Tree-shape OOD (STRUCTURE-AGNOSTIC)

Cell trained on BALANCED RPN, applied to LINEAR RPN *without retraining*:

Depth	1	2	3	4	5	6	7	8
Accuracy	1.00	1.00	1.00	1.00	0.994	0.974	0.956	0.928

$d = 1..4$ perfect. Pequeña degradación a $d \geq 5$ es drift numérico de aplicaciones recursivas. Cell es **structure-agnostic**.

Gate 1: Boolean formulas — principio generaliza

Boolean RPN, vocab {0, 1, AND, OR, XOR}, chance = 50%:

arch	d2	d3	d4	d5	d6	d7	d8
TreeMoTE	1.00	1.00	1.00	1.00	1.00	1.00	1.00
StackMoTE	1.00	1.00	1.00	1.00	0.95	0.81	0.79
Transformer	1.00	0.99	0.98	0.95	0.72	0.74	0.71

TreeMoTE = 100% en **task family distinta** a mod-arithmetic. Principio generaliza.

Gate 2: TreeMLP control — principio es recursive-sharing, no typed ops

Mod-7	d2	d3	d4	d5	d6	d7	d8
TreeMLP	1.00	1.00	0.996	0.994	0.992	0.988	0.972
TreeMoTE	1.00	1.00	1.00	1.00	1.00	1.00	1.00

TreeMLP también 100% a d=8. Principio es **tree-recursive shared-cell + parse-given**, no TreeMoTE específico. Cell parameterization afecta convergence speed pero no structural elimination.

Gate 3: Cell audit en ambos dominios

- Mod-7: TreeMoTE aprendió 147/147 (a, b, op) triples correctos.
- Boolean: 12/12 {AND, OR, XOR} triples correctos.

White-box mechanistic proof: cell learns *exact* operator table.

Gate 5: Parse corruption (causal evidence)

Load trained TreeMoTE, shuffle operand positions at test with rate c :

depth	$c=0\%$	$c=10\%$	$c=25\%$	$c=50\%$	$c=100\%$
2	1.000	0.934	0.872	0.748	0.528
3	1.000	0.926	0.844	0.702	0.356
5	1.000	0.910	0.784	0.594	0.190
8	0.996	0.892	0.808	0.632	0.158

At full corruption, $d = 8$ accuracy collapses to 15.8% (chance = 14.3%). **Parse es causalmente necesario.**

Gate 6: Coverage Law

Multi- p scaling, 15K steps, curriculum {2, 3, 4, 5}:

p	coverage	TreeMoTE $d = 8$	TreeMLP $d = 8$	Transformer $d = 8$
7	100%	1.000	0.972	0.174
11	25%	0.992	0.172	0.114
17	7%	0.226	0.072	0.060
31	18%	0.104	—	0.034

Findings:

- TreeMoTE siempre $>$ TreeMLP $>$ Transformer.
- Con coverage $\geq 25\%$, TreeMoTE $\rightarrow 100\%$.
- TreeMLP requiere *full* coverage para funcionar (falla a 25%).
- Typed bilinear ops (TreeMoTE) son **sample-efficient** bajo sparse coverage.

Claim final refinado:

- Recursive shared-cell + parse-given $\Rightarrow$ elimina depth decay (structural; cualquier cell tree-recursive funciona).
- Typed bilinear cell (TreeMoTE) $\Rightarrow$ sample-efficient operator learning bajo partial coverage.
- Dos axes ortogonales. Flat architectures fallan en ambos.

Estado final: DIAMANTE sólido

Para NETHERITE faltan:

1. Real task (Lean4 AST, Code AST, GSM8k structured)
2. Parse-then-evaluate two-stage model
3. TPU-scale experiments ($p = 97+$, $d_{\text{model}} = 512+$)

Test 4: $p=31$ (larger modulus)

arch	d2 (train)	d3 (train)	d4 (OOD)	d5 (OOD)
TreeMoTE	0.526	0.410	0.266	0.180
Transformer	0.118	0.072	0.030	0.052

Chance = $1/31 = 0.032$.

TreeMoTE NO convergió completamente (coverage 18% insuficiente), **pero gap arquitectural persiste**:

- TreeMoTE d5 OOD = $5.6\times$ chance
- Transformer d5 = $1.6\times$ chance
- Gap absoluto +13pp, relativo $3.5\times$

3 Dispatch-tax hypothesis — validación dual

La hipótesis se confirma en dos regímenes distintos:

- $p=7$, $n=3000$ (full coverage): Tree 100% vs Trans 17% $\rightarrow$ **+83pp gap**
- $p=31$, $n=50K$ (18% coverage): Tree 18% vs Trans 5% $\rightarrow$ **+13pp gap, $3.5\times$ relative**

En ambos, TreeMoTE > Transformer por factor $> 3\times$. Claim arquitectural robusto, independiente de cell-learning convergence.

4 Reinterpretación del breakthrough

El TreeMoTE breakthrough es **CONDICIONAL**:

- TreeMoTE resuelve **depth extrapolation** ($\alpha \rightarrow 0$) IFF cell learning succeeds.
- Cell learning es un problema estándar de data + compute, separado del structure problem.

Claim precise: “Tree-recursive arch eliminates the depth-decay tax when operator is learnable.”

Claim NO establecido: “TreeMoTE resuelve compositional generalization universally” — need more data/compute for larger p .

5 Key findings consolidados

5.1 La ley empírica α (pre-TreeMoTE)

$$\text{acc}(d) = \text{floor}_{\text{arch}} + (\text{acc}(d_{\min}) - \text{floor}_{\text{arch}}) \cdot e^{-\alpha_{\text{arch}}(d-d_{\min})}$$

arch	floor	α
Transformer	0.16	0.78 ± 0.18
MoTE	0.16	0.76 ± 0.05
StackMoTE	0.22	0.42 ± 0.09
TreeMoTE	≈ 1.0 (for p=7)	≈ 0

MAE prediction OOD (fit on $d = \{2, 3, 4\}$, predict $d = 5, 6, 7, 8$): 1.7–3.2pp.

5.2 Architecture hierarchy emergente

Arch	α	Peak OOD (d=8)
Transformer	0.78	17%
MoTE	0.76	17%
StackMoTE	0.42	20%
TreeMoTE	≈ 0	99%+

6 Retracciones / correcciones (disciplina científica)

1. **Null model error (2026-04-24 mañana):** Claim original “+7-8pp chance” era “+1-3pp majority-class”. Corregido con rejection-sampled uniform labels.
2. **Single-seed hype (2026-04-24 tarde):** d128_L2 balanced +16pp no reproducía. Steady state 23%. Retraído.
3. **“MoBE gana math LM” (2026-04-23):** Transformer 4.5% mejor en 11 evals. Claim muerto.

7 Limitaciones conocidas

- Max params $\leq 3.2\text{M}$ en Transformer/Stack; TreeMoTE testado a $\sim 65\text{K}$
- Solo balanced RPN mod-7 y mod-31
- Max 50K steps
- TreeMoTE requires tree structure explicitly (via token positions)
- No testado en natural language ni tasks con structure ambigua

8 Implicaciones prácticas

Para cualquier task composicional con estructura explícita:

- Si tiene parse conocido (Lean4, SQL, AST, math expressions) $\rightarrow$ construir arch tree-recursive ($\alpha = 0$)
- Si no (natural language) $\rightarrow$ tenés que pagar el dispatch tax. Ley α aplica.

Design criterion: separate parsing from evaluation. Don’t conflate them architecturally.

9 Estado de archivos

Paper:

- `paper/compositional_depth_law.tex + .pdf` — v4. Needs v5 rewrite incorporando breakthrough.

Raw data:

- `raw_data/_scratch/` — 33+ results.json + 4 param pickles + mass_entropy_results.json + tree_mote_test + tree_stress + tree_p31_coverage + matched_params

Code (repo local + remoto):

- `_kazdov_jax/kazdov_jax/stack_mote.py`
- `_kazdov_jax/kazdov_jax/tree_mote.py` — **NEW 2026-04-24**
- `_kazdov_jax/scripts/stack_mote_comparison.py` — driver con 7 archs soportadas

10 Infrastructure

- RTX 4090 remote via Tailscale (100.80.106.61)
- WSL2 Ubuntu, JAX con CUDA
- Google TRC TPU: v4 32 chips on-demand + v6e 64 chips spot (Trillium), 29 días restantes

11 Plan forward

Inmediato (4090, en progreso)

1. Matched-params mini head-to-head (Tree vs Trans vs Stack at $\sim 30K$ params) — running
2. Paper v5 rewrite con breakthrough + caveats honestos
3. BITACORA compilada a PDF — este documento

Fase 2 (4090 o TPU)

1. $p=31$ con $n=200K$ (full coverage) — confirmar breakthrough a p mayor
2. Linear RPN variant para generality
3. arXiv preprint para timestamp

Fase 3 (TPU, próximos días)

1. GRAN modelo (`d_model=512`, $\sim 20M$ params)
2. $p=97$ o $p=197$ test
3. Multi-seed tight error bars (10+ seeds)
4. Depth extrapolation to $d = 20+$
5. Lean4 tactic composition si arch permite

Fase 4 (futuro)

1. Aplicar insights al 5B math pipeline
2. Hybrid LLM + TreeMoTE para math reasoning real
3. Paper 2: real-world math tasks

12 Competencia

- **Memory Caching** (Google, Behrouz et al. 2026) — aborda long-context recall. Orthogonal.
- **MesaNet** (Google, ICLR 2026) — in-context learning linear layers. Orthogonal.
- **AAAI 2026 paper 2512.02677** — LLM recursive logic fails. Task-different.
- **Petty et al. TMLR 2025** — depth effect compositional gen. Doesn't derive α law.

Ninguno tiene: la ley α cuantitativa, el dispatch tax mechanistic claim, TreeMoTE breakthrough en depth extrapolation. Ventana de publicación abierta.

Nota personal

La ansiedad de presentar en conferencias no debe bloquear la publicación. Opciones válidas:

- arXiv-only (80% del impacto vive en arXiv + tweet, no en conference talk)
- Co-autor o colaborador presenta
- Video pregrabado (NeurIPS/ICLR aceptan)
- Workshop con poster (menos stage)

El paper puede salir sin hablar en público.